



Quectel EC200N-CN Mini PCIe-C LTE Standard Module

Introduction

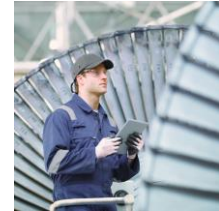
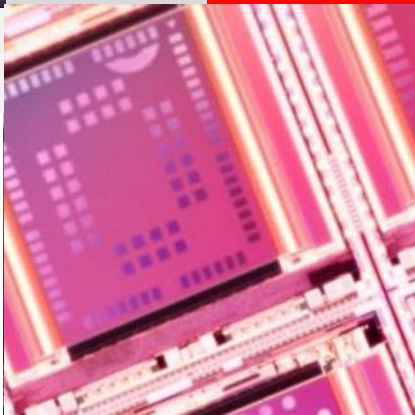
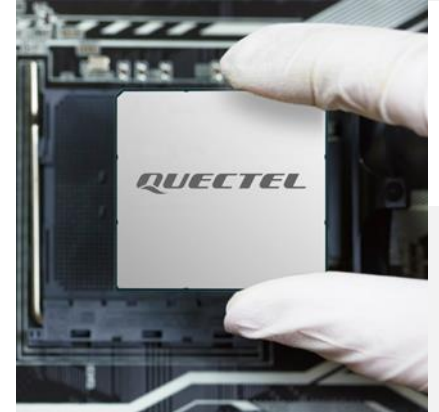
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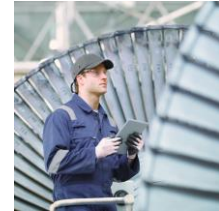
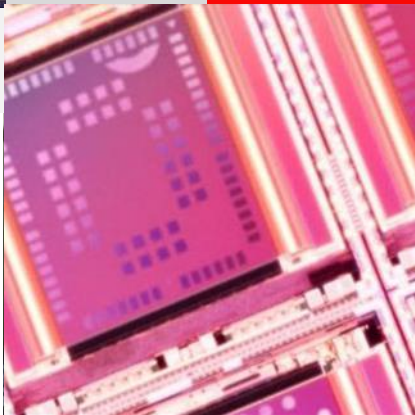
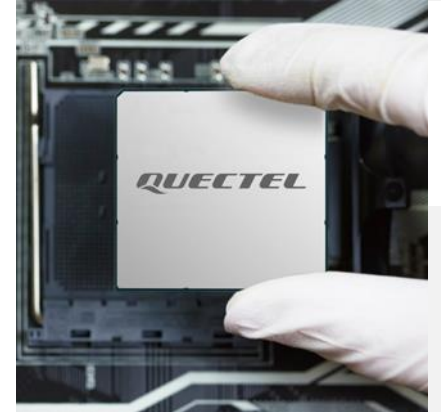
Highlights & Specifications

Development Timeline

Technical Details

Applications

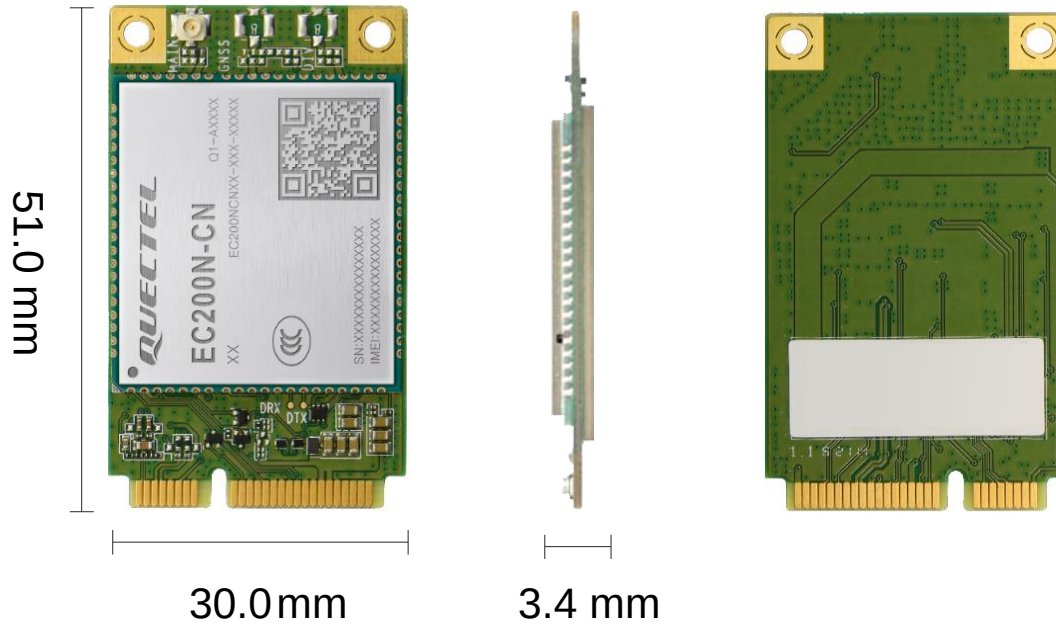
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EC200N-CN Mini PCIe-C Mechanical Dimensions



LTE Cat 1 Module (ASR1603)



Length: 51.0 mm (± 0.15 mm)
Width: 30.0 mm (± 0.15 mm)
Height: 3.4 mm (± 0.2 mm)
Weight: Approx. 11.4 g

EC200N-CN Mini PCIe-C Specifications



LTE Cat 1 Module

51.0 mm × 30.0 mm × 3.4 mm
10 Mbps DL/ 5 Mbps UL

Module		EC200N-CN Mini PCIe-C
LTE	LTE-FDD	B1/ B3/ B5/ B8
	LTE-TDD	B34/ B38/ B39/ B40/ B41
GSM		-
QuecLocator®		Supported
Analog Audio		Supported
VoLTE		Supported
(U)SIM Detection		Supported
DFOTA		Supported
Form Factor		Mini PCIe
Region		China/ India
Size (mm)		51.0 × 30.0 × 3.4
Certification		Carrier: China Telecom* Regulatory: SRRC/ NAL/ CCC

“*” means ongoing.

EC200N-CN Mini PCIe-C Main Highlights



Compatible

- The promotion area and the supported LTE frequency bands of EC200N-CN Mini PCIe-C are the same as that of EC200S-CN Mini PCIe-C.

Performance-optimized

- Higher integration by using SIP Flash solution.
- Highly price-competitive.

Reliable

- Meet the standards of the IoT industry and related regional certifications.
- Conduct detailed internal tests to meet industry standards for industrial-grade products.



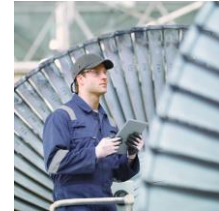
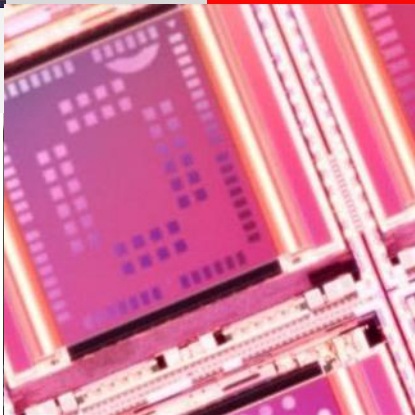
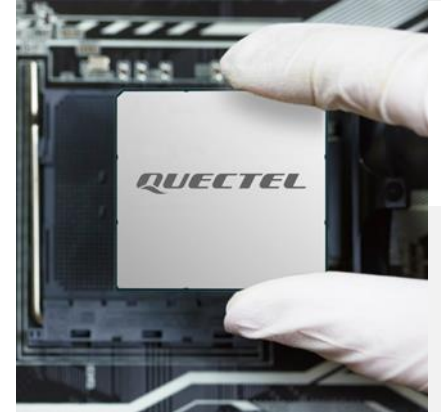
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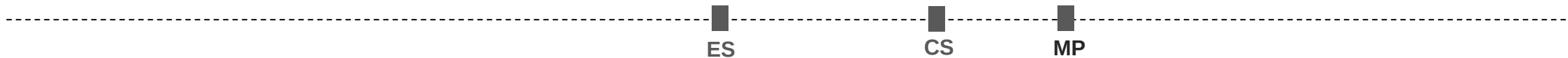


EC200N-CN Mini PCIe-C Timeline



2021											
Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.

Project Schedule



ES: Engineering samples ready. Basic functions are available for customers' simple demo purpose.
CS: Commercial samples ready. Stable hardware design and quite stable software design. New software features can be added upon request.
MP: Hardware and software ready for mass production. For certification status, please refer to the certification schedule below.

Carrier Certification



Regulatory Certification

SRRC/ NAL/ CCC



Others

WHQL



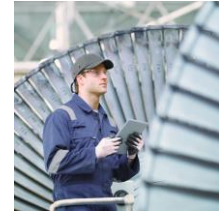
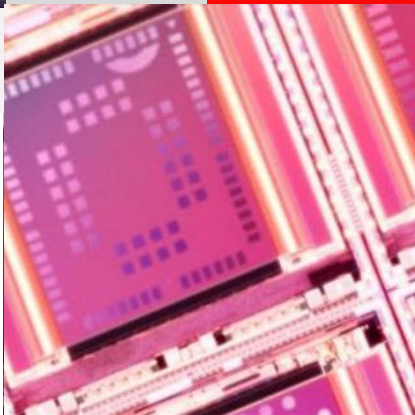
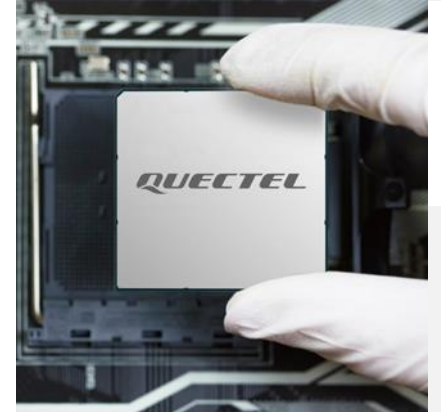
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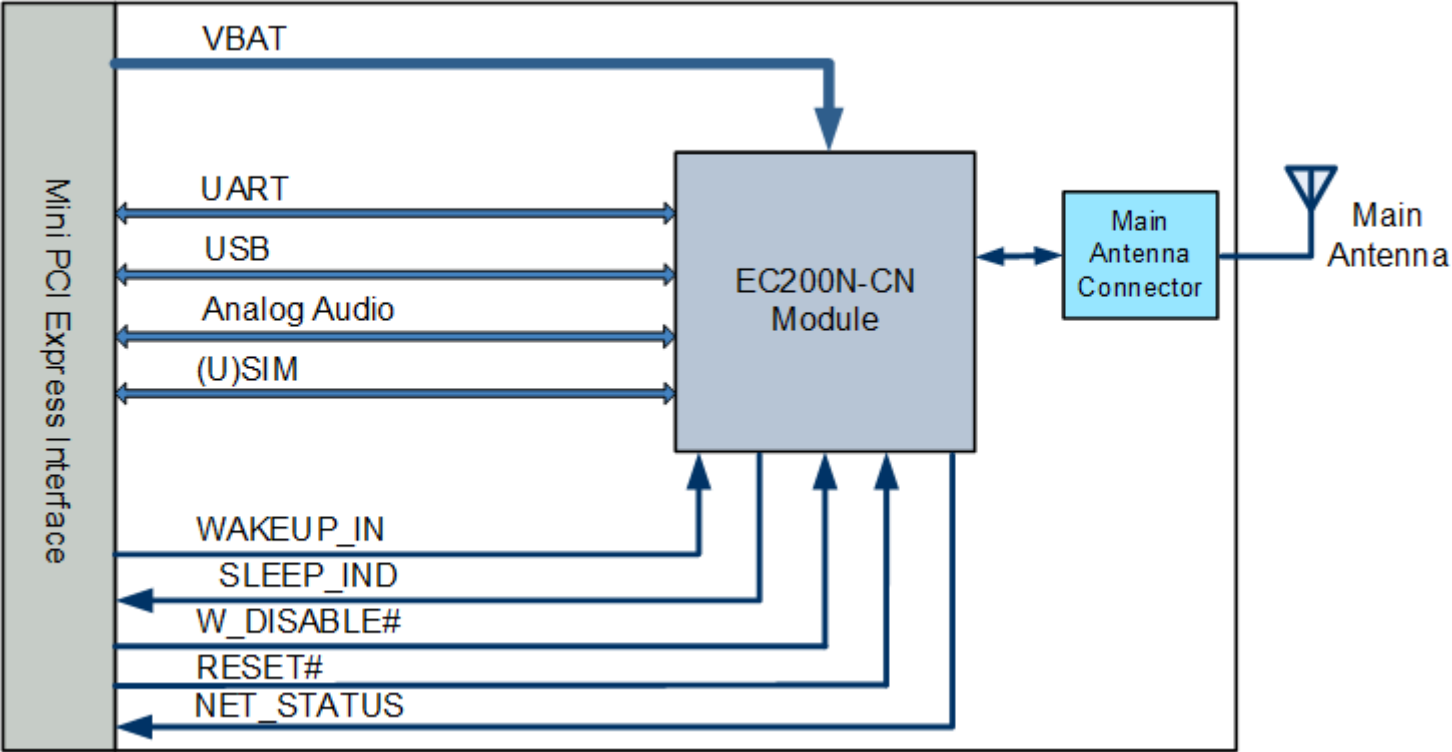
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EC200N-CN Mini PCIe Hardware Architecture



Software Advantages



USB Serial Driver

- Windows 7/8/8.1/10
- Linux 2.6–5.12
- Android 4.x–11.x

Other Drivers

- RIL Driver: Android 4.x–11.x
- USB RNDIS Driver:
Windows 7/8/8.1/10
Linux 2.6–5.12
- USB ECM Driver:
Linux 2.6–5.12

Quality Guarantee

- Reliable network protocols
- Steady flash protection mechanism
- Superior audio algorithms

Special Features

- DFOTA
- QuecFile®

Flexible Applications

- (U)SIM card detection
- DTMF
- TTS

Enhanced AT Commands

- 3GPP TS 27.007 (GSM 07.07)
- 3GPP TS 27.005 (GSM 07.05 SMS)
- Quectel enhanced AT commands

DFOTA

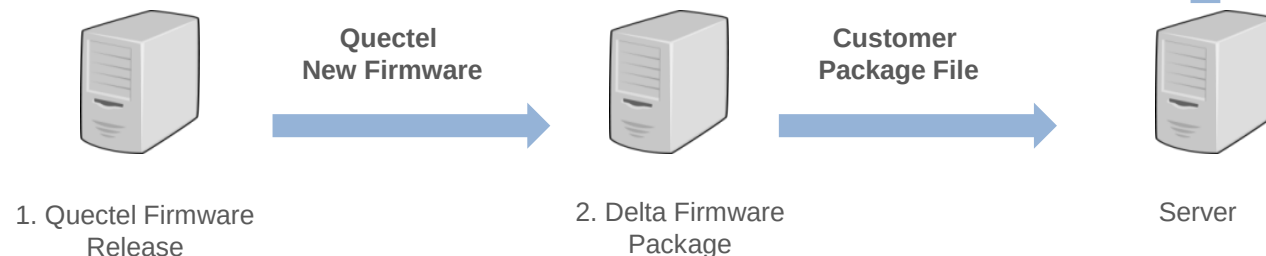


DFOTA refers to Delta Firmware Upgrade Over-The-Air, which enables customers to upgrade firmware wirelessly.

In DFOTA, a delta firmware package, which only contains the differences between the source and the target firmware versions, occupies small memory. Thus the firmware can be upgraded quickly over LTE networks, eliminating the need for users to bring the device to a service facility.

Firmware Upgrade Process via DFOTA

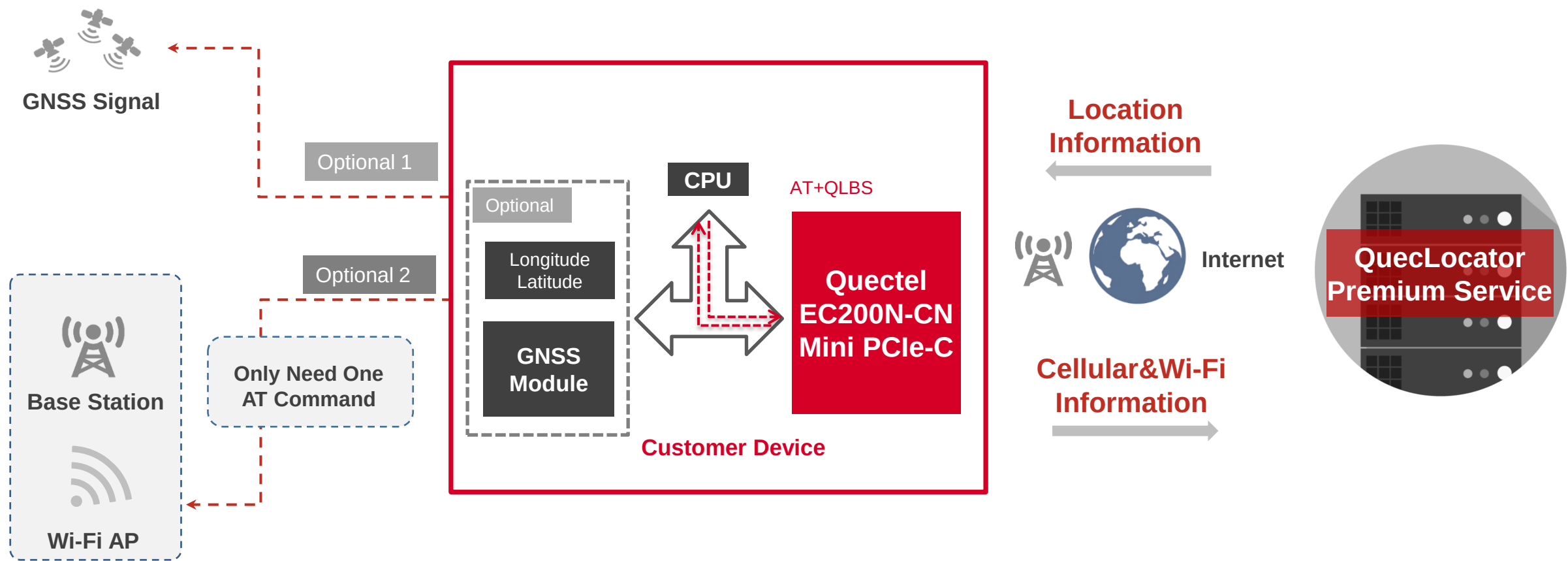
- Get Delta Firmware Package
- Put Delta Package on Server
- Execute **AT+QFOTADL** Command
- Download and Upgrade Firmware Automatically



QuecLocator[®] Working Process



QuecLocator[®] is an easy and reliable positioning service providing secure and accurate positioning **Anytime, Anywhere.**



NOTE: There is no need for extra GNSS modules when using QuecLocator[®].

Mini PCIe-C Technical Support Package

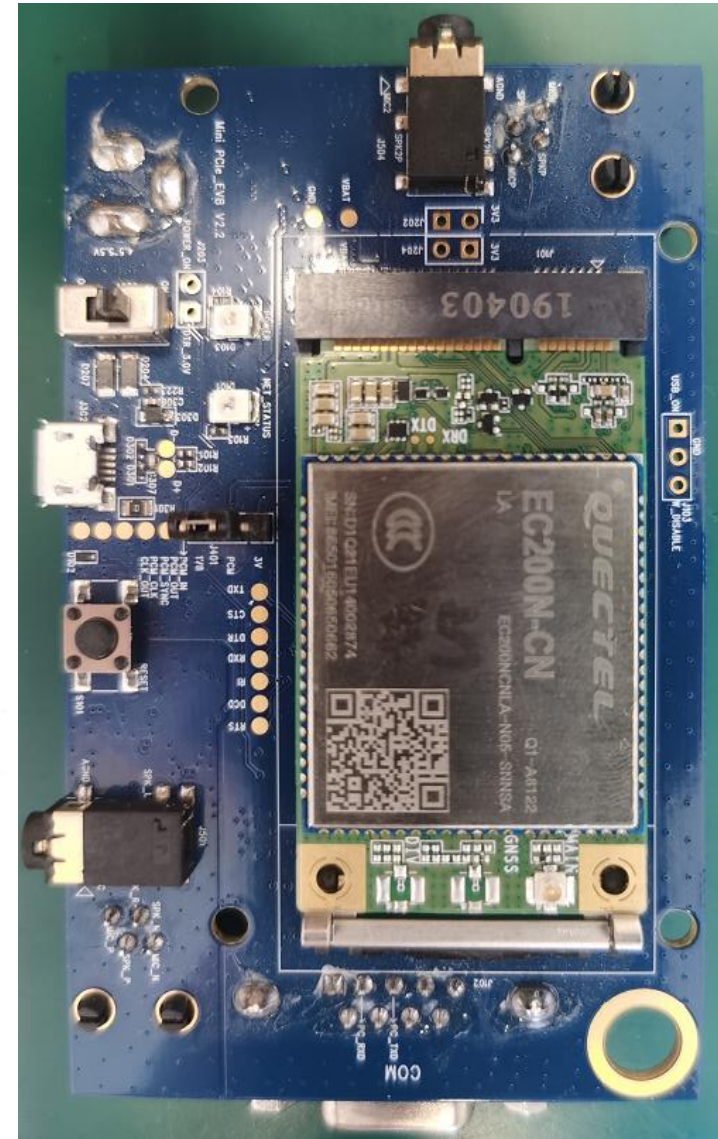


Technical Materials Package

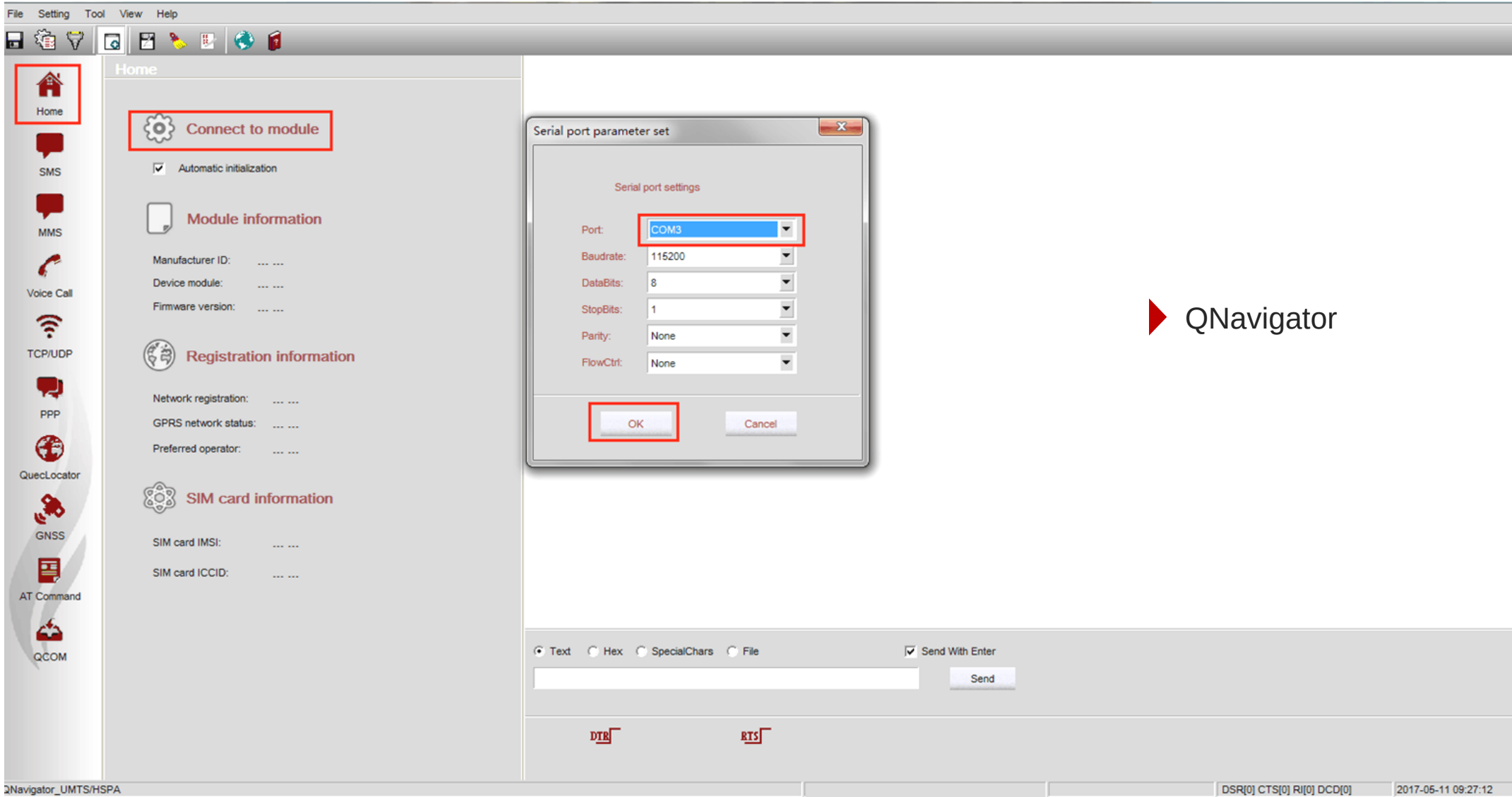
- Hardware & software documents
- Debug tools, download tools, test tools, EVB kit & USB drivers
- Approvals & test reports
- QNavigator

Development Tool (Mini PCIe-C EVB Kit)

- | | |
|---------------------------------|-----------------------|
| ▪ Connectors | ▪ Auxiliaries |
| a) Power supply connectors | a) Network status LED |
| b) USB connector | b) Power switch |
| c) UART connectors | c) Reset key |
| d) (U)SIM card connector | d) Test points |
| e) Digital and audio connectors | |



Support Package - QNavigator



► QNavigator

Technical Support Team





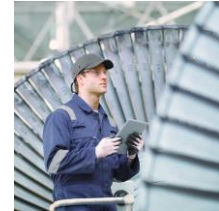
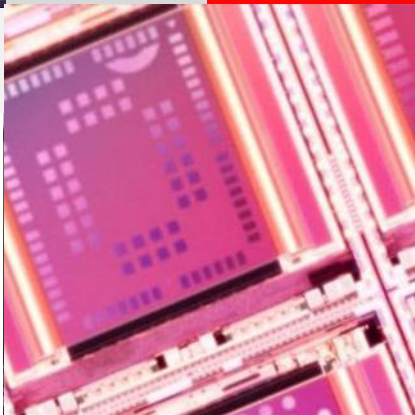
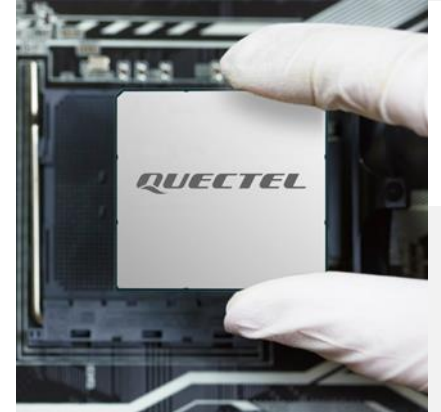
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Target Applications



Telematics & Transport

- Automotive OEM
- Vehicle Tracking
- Container Tracking
- Ship Tracking
- Fleet Management
- OBD
- DVR
- Insurance Box



Energy

- Electricity Meter
- Gas Meter
- Water Meter
- Heat Meter
- Smart Grid
- Wind Turbines
- Solar Panel
- Charging Pile



Payment

- Wireless POS
- Cash Register
- ATM
- Vending Machine
- Top-up Machine



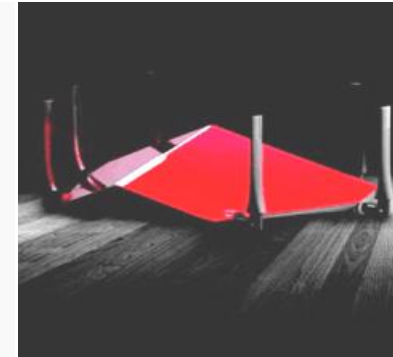
Security

- Alarm
- Camera
- Video Surveillance
- Intrusion Detection
- Smoke Detector
- Asset Protection



Smart City

- Street Lighting
- Sharing Economy
- Traffic Light
- Elevator
- Digital Signage
- Advertisement Display
- LED Lighting
- Garbage Bin Monitoring
- Parking



Gateway

- DTU
- Consumer Router
- Industrial Router
- VoIP
- Server
- Wi-Fi Hotspot



Industry

- Industrial PDA
- Rugged Tablet PC
- Pipeline Management
- UAV
- Robot
- Flow Meter
- Refrigerator
- Industrial Control
- Industrial Monitoring



Life & Healthcare

- Personal Tracker
- Household Appliances
- Pet Tracker
- Wearables
- Elderly Monitoring
- Remote Medical Equipment
- Glucometer
- Blood Pressure Monitor
- Gambling Machine



Agriculture & Environment

- Trail Camera
- Food Traceability
- Farm Machinery
- Irrigation
- Meteorological Station
- Wildlife Tracking
- Environment Monitoring
- Pollution Monitoring



The number one cellular module vendor in the world and a leading GNSS module supplier

- Unbeatable choice from the broadest module portfolio in the world
- The highest quality products for the best possible prices
- Superb support with the largest R&D team in the industry
- Continuous innovation – first to market with 5G, LPWA, CV2X, snapdragon
- A passionate, dedicated team of “Quectelers” ensure our customers always come first

Thank You

Build a Smarter World

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